

EZHILVENTHAN KULANDAIVEL

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SUMMARY

Data Science graduate student at the University of Alabama at Birmingham with hands-on experience in machine learning, artificial intelligence, predictive analytics, and data visualization. Skilled in Python, SQL, statistical analysis, and end-to-end data science workflows, including data preparation, feature engineering, model development, and performance evaluation. Experienced in building AI-driven solutions through projects involving computer vision, anomaly detection, recommendation systems, and interactive dashboards. Passionate about developing responsible AI solutions that solve complex business problems and create measurable impact.

EDUCATION

University of Alabama at Birmingham	Jan 2025 - Dec 2026
Master's in Data Science(Current) GPA: 3.2	
Dr. M.G.R Educational And Research Institute	Sep 2021- Sep 2024
Bachelor of Computer Application in Artificial Intelligence & Data Science, GPA: 7.9/10	

ACADEMIC PROJECTS

E-Commerce Sales Analytics Dashboard:	Jun 2026 - Jun 2026
- Developed an interactive Excel dashboard to analyze sales performance, customer behavior, product trends, and delivery metrics. - Created KPI reporting, pivot table analyses, and visualizations to support business decision-making. - Utilized data cleaning, transformation, and dashboard design techniques to present actionable reports.	
Traffic Monitoring Drone System:	Feb 2026 - Apr 2026
- Built a real-time traffic analytics platform that processes drone footage using YOLOv8 and OpenCV to detect, track, and count vehicles. - Generated traffic insights including vehicle volume, congestion levels, and illegal turn detection to support traffic management decisions. - Designed a MySQL database pipeline for storing and analyzing traffic and violation data. - Developed a JavaFX dashboard with visual reports and analytics for monitoring traffic patterns and system performance. - Integrated AI models with backend services using FastAPI, enabling efficient data flow between detection, storage, and visualization.	
Credit Card Anomaly Detection:	Feb 2025 - Apr 2025
- Developed an anomaly detection framework for identifying potentially fraudulent credit card transactions. - Applied PCA and SVD to reduce dimensionality and improve pattern discovery in high-dimensional financial data. - Implemented K-Means Clustering to separate normal transaction behavior from anomalies. - Evaluated unsupervised learning techniques for fraud detection and risk analysis.	
Data Analysis Chatbot:	Dec 2023 - Apr 2024
- Created a data analysis robot assisting people in viewing and comprehending data in a simple manner - Developed the program using Python programming language along with the Streamlit library for creating an interactive interface. - Data cleaning and manipulation with Pandas - Emphasized ease of use in analyzing data, allowing users who may not be tech-savvy to quickly adapt.	

TECHNICAL SKILLS

- Excel (Pivot Tables, Dashboards, Data Validation, Data Cleaning)
- SQL
- Power BI
- Python (Pandas, NumPy, Matplotlib)
- Data Analysis & Reporting
- Data Quality Verification
- Data Visualization
- Data Management
- KPI Development
- Trend Analysis

CERTIFICATIONS (COGNITIVE CLASS.ai by IBM)

Certified in Introduction to Python course	June 2023 - July 2023
Gained a strong foundation in Python programming through structured coursework and hands-on practice.	
Certified in Data Visualization course	June 2023 - July 2023
Developed practical knowledge of data visualization by learning how to present data using various charts and graphical formats.	
Certified in Cloud Application course	June 2023 - July 2023
Learned the basics of cloud computing and the fundamentals of cloud-based applications.	